

Functions

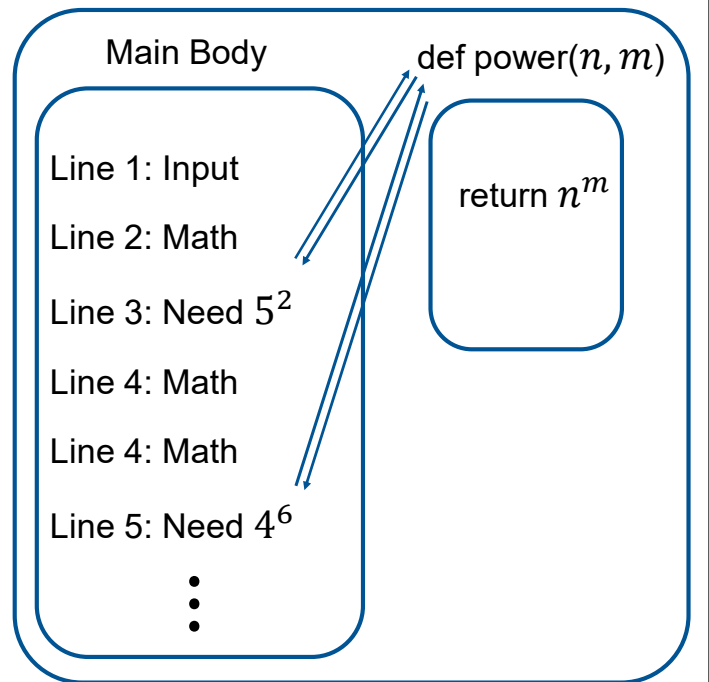
What is a function?

A separate part of your code (normally at the top).

A self-contained block of code that performs a specific task.



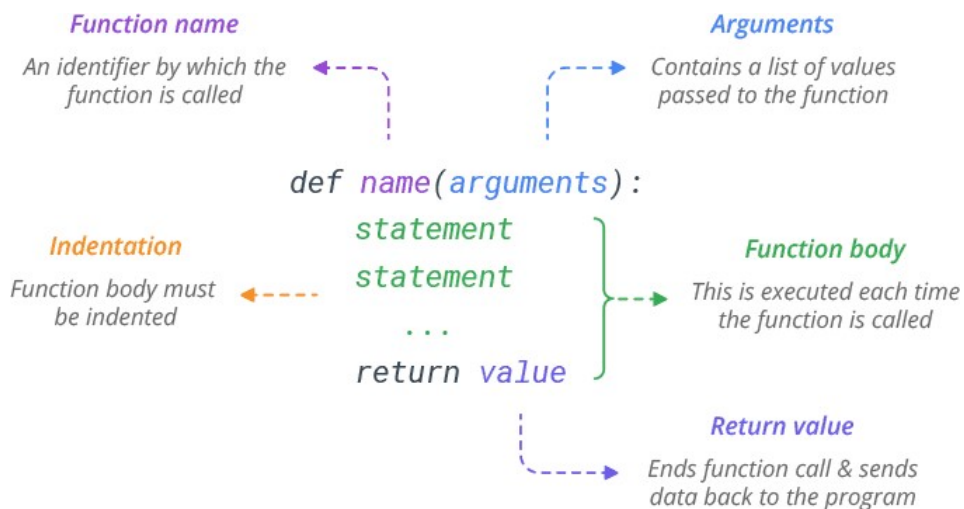
Your Code



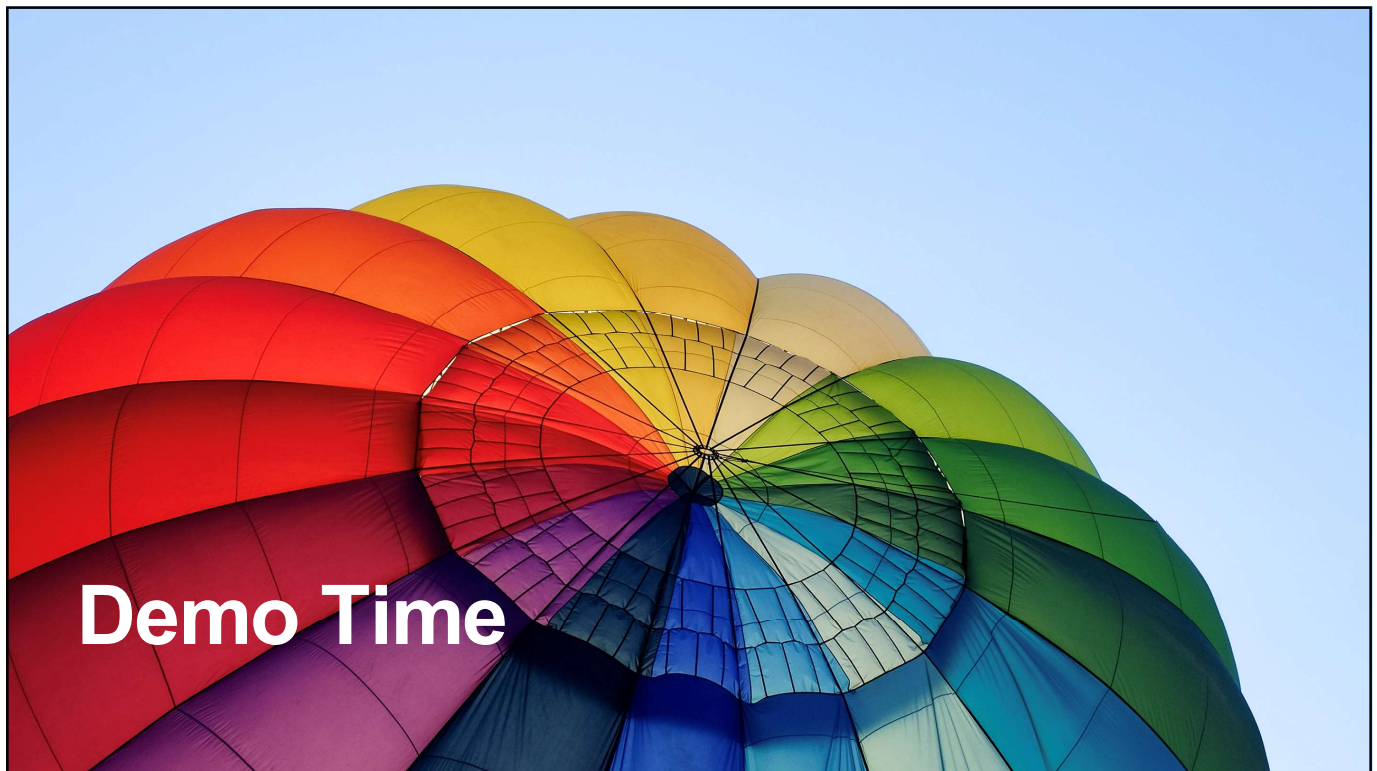
1

Functions

Basic syntax of a function in Python.



2



3

Before We Go

if `__name__` == "`__main__`"

- In Python, the `if __name__ == "__main__"` construct is used to determine if a Python file is being run directly or if it is being imported as a module in another file. `__name__` Variable:
- Every Python script has a special built-in variable called `__name__`.
- If a script is run directly, e.g. `python script.py`, Python sets `__name__` to "`__main__`".
- If a script is imported as a module in another script, `__name__` is set to the name of that script instead.
- `if __name__ == "__main__"` makes sure that the code inside this block will only execute if the script is run directly, not if it is imported elsewhere.

4

CW3

Keep calm and code on ;-)

5

Before We Start

I have a bad memory so this is what I do in classes.

- Take note in as much detail as I can.
- After class, rewrite and organise my note, so it is in better handwriting and I can add things that I didn't have time to write down during class.
- just take a picture of an interesting slide.

6

Required from SW2

So these things you should know how to do.

- Get inputs from the user.
- Calculate the overall_score.
- Find the nearest available grade.

7

CW3: The Requirements

These are the libraries that we need.

- The `math` library for our calculations
- the `datetime` library for calculations of age.
- The `tabulate` library for creating my table.

Just in case, this is how I import my libraries:

```
import math
from datetime import datetime
from tabulate import tabulate
```

8

Defining Functions

We need to define some functions in our code

- The list of functions are given in the coursework specification.
- The functionality of each is obvious from their names.
- All function should have a return command.
- There is also a main function.
 - The correct use of main function was just taught.

9

Going Through an Example Code

Let us see an example of a code that passes all tests.

- Executing the code first gives a welcome message (not really necessary).

```
Welcome to the Student Grading System First:  
Let us set up the module configuration.
```

- Then, it will ask if I would like to configure a new module.
- I will say no for now.

```
Do you want to configure a module? (y/n):
```

```
n|
```

10

Going Through an Example Code

Let us see an example of a code that passes all tests.

- This prompts another question regarding automatic or manual data entry.
- I will select manual for now.

```
Do you want to (1) input manually or (2) load from file?
1|
```

- Now I can enter the data for a maximum of three students.
- Entering “end” should end the process and create the required “*.txt” file.
- Let us see how this works.

11

Going Through an Example Code

Let us see an example of a code that passes all tests.

```
Please enter student ID (2-digit number or 'end' to finish): 12
Please enter student name: Mark
Please enter student's date of birth (YYYY-MM-DD): 2000-12-03
Please input the student's grade for assessment component 1: 100
Please input the student's grade for assessment component 2: 45
Please input the student's grade for assessment component 3: 45
Please input the student's grade for assessment component 4: 45

Please enter student ID (2-digit number or 'end' to finish):
|
```

12

Going Through an Example Code

- Now let us see how it goes for configuring a new course.

```
Welcome to the Student Grading System First:
Let us set up the module configuration.
Do you want to configure a module? (y/n): y
Enter module name: SD1-2023
How many assessment components does SD1-2023 have? 3
Component 1 name: CW1
Component 1 weight (%): 20
Component 2 name: CW3
Component 2 weight (%): 40
Component 3 name: Exam
Component 3 weight (%): 40
Module configuration complete.
Do you want to (1) input manually or (2) load from file? 1
Please enter student ID (2-digit number or 'end' to finish): 12
Please enter student name: Mark
Please enter student's date of birth (YYYY-MM-DD): 2000-12-03
Please input the student's grade for assessment component 1: 100
Please input the student's grade for assessment component 2: 80
Please input the student's grade for assessment component 3: 45
Please enter student ID (2-digit number or 'end' to finish):
```

13

Going Through an Example Code

Let us see an example of a code that passes all tests.

- Now what happens if I want energies to be done automatically.

```
Welcome to the Student Grading System First:
Let us set up the module configuration.
Do you want to configure a module? (y/n): n
Do you want to (1) input manually or (2) load from file? 2
```

Enter the filename to load student data from:

- The code accepts a file txt file name and inputs student data from there.
- Then, press the student data and saves the results in a separate text file.

14

Final Note

- You should be able to interoperate CodeGrade Feedback.
- Please submit any code that you have for CW3 to CodeGrade.
- See the feedback and ask questions.

